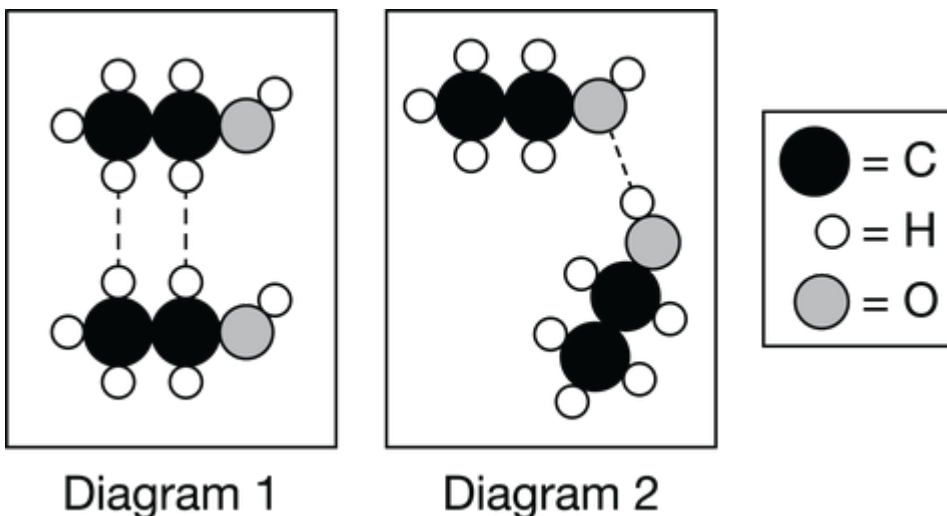


Unit 3 Progress Check: MCQ

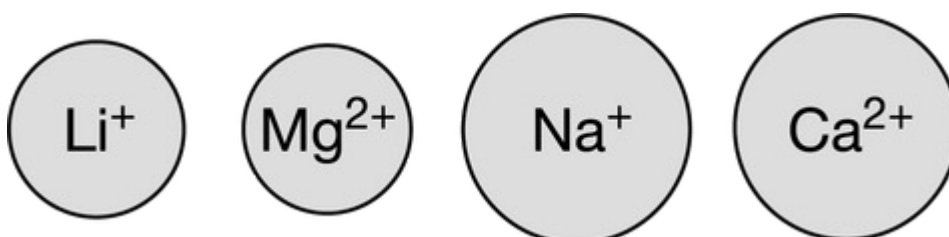
1.



Which particle diagram shown above best represents the strongest intermolecular force between two ethanol, $\text{C}_2\text{H}_6\text{O}$, molecules?

- (A) Diagram 1, because it shows hydrogen bonds forming between hydrogen atoms from different ethanol molecules.
- (B) Diagram 1, because it shows strong, directional dipole-dipole forces between two polar ethanol molecules.
- (C) Diagram 2, because it shows the formation of a hydrogen bond between an H atom bonded to an O atom with an O atom from another molecule. ✓
- (D) Diagram 2, because it shows the dipole from an ethanol molecule inducing a dipole in another ethanol molecule.

2.



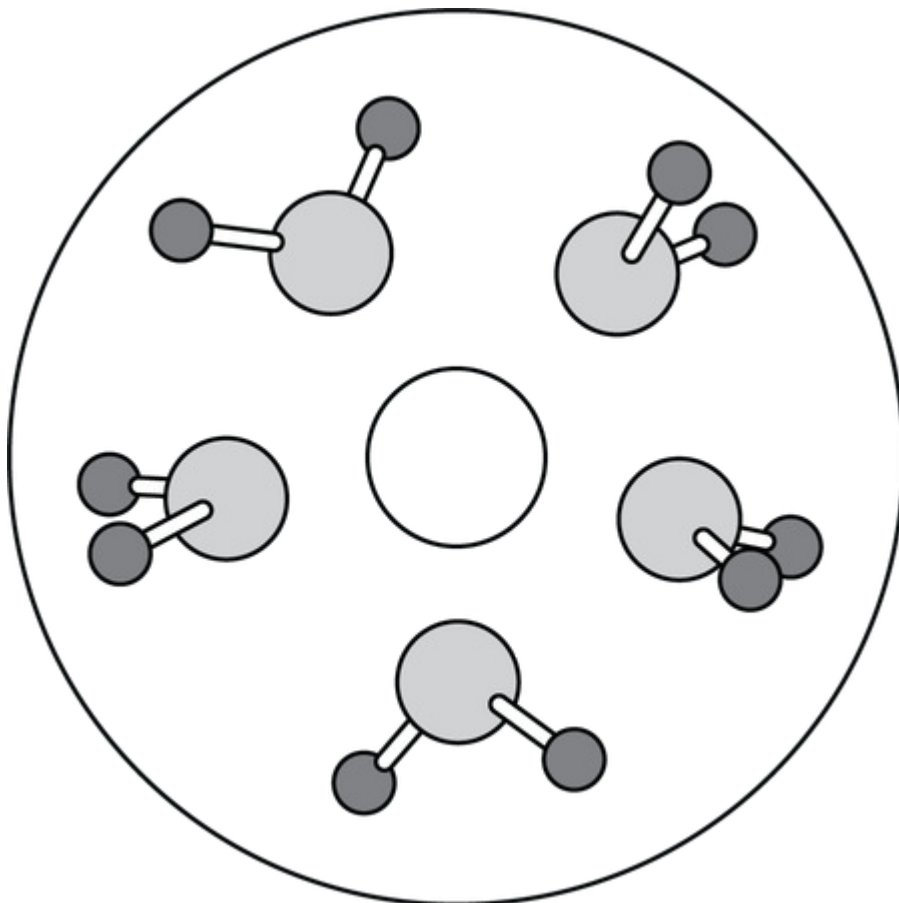
The diagram above represents four cations, all shown to the same scale. Which cation would be predicted by Coulomb's law to have the strongest ion-dipole attraction to water, and why?



Unit 3 Progress Check: MCQ

- (A) Li^+ , because it is the smallest group 1 metal ion.
- (B) Mg^{2+} , because it has the largest charge-to-size ratio. ✓
- (C) Na^+ , because it has the smallest charge-to-size ratio.
- (D) Ca^{2+} , because it is the largest group 2 metal ion.

3.



A solid compound of a group 1 (alkali) metal and a group 17 (halogen) element dissolves in water. The diagram above represents one type of solute particle present in the solution. Which of the following identifies the solute particle and best helps explain how the solute particle interacts with water molecules?



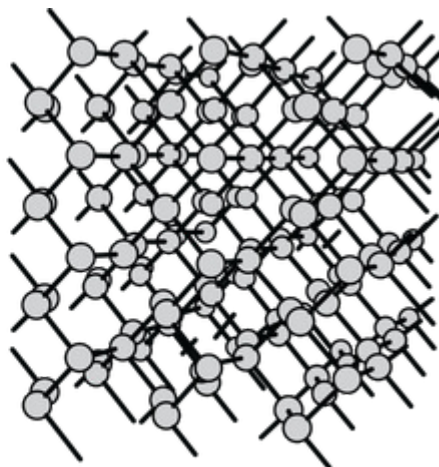
Unit 3 Progress Check: MCQ

- (A) The particle is a negative ion, and the interactions are hydrogen bonds.
- (B) The particle is a negative ion, and the interactions are ion-dipole attractions.
- (C) The particle is a positive ion, and the interactions are ion-dipole attractions. ✓
- (D) The particle is a positive ion, and the interactions are dipole-dipole attractions.

4.



Allotrope I



Allotrope II

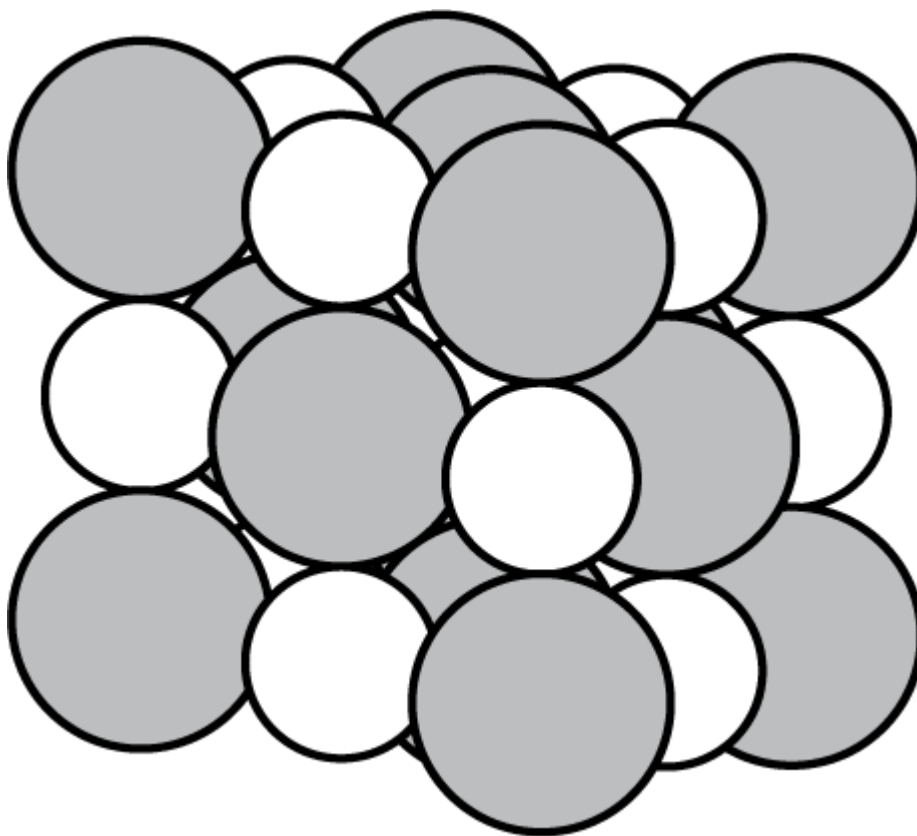
The diagrams above represent two allotropes of solid phosphorus. Which of the following correctly identifies the allotrope with the higher melting point and explains why?

- (A) Allotrope I, because it has P_4 molecules with covalent bonds that are stronger than dispersion forces between individual P atoms in allotrope II.
- (B) Allotrope II, because it has covalent bonds between the phosphorous atoms that are stronger than the dispersion forces between the P_4 molecules in allotrope I. ✓
- (C) Allotrope II, because it has metallic bonding that is stronger than the covalent bonding found in the P_4 molecules in allotrope I.
- (D) Both allotropes have the same melting point because they both contain phosphorous atoms.



Unit 3 Progress Check: MCQ

5.



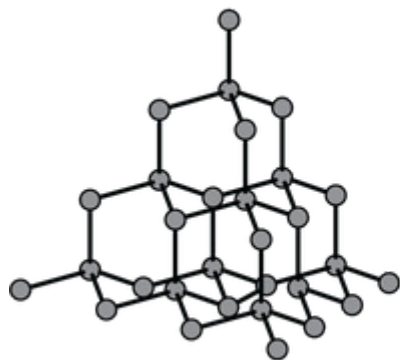
The crystal structure of **NaBr** is represented in the diagram above. Which statement correctly compares crystalline **NaBr(s)** to molten **NaBr(l)** in terms of electrical conductivity?

- (A) Crystalline **NaBr** contains no freely moving electrons that could conduct an electrical current, whereas electrons can flow freely in molten **NaBr**, which is a good conductor of electricity.
- (B) Crystalline **NaBr** and molten **NaBr** both contain ions that are held in fixed positions due to strong electrostatic attractions among the ions, making neither a good electrical conductor.
- (C) Crystalline **NaBr** and molten **NaBr** both contain **Na** atoms that transfer electrons to **Br** atoms in a chemical reaction, thus allowing them both to be good conductors of electricity.
- (D) Crystalline **NaBr** contains no freely moving electrons to conduct electricity, but molten **NaBr** is composed of freely moving **Na⁺** and **Br⁻** ions, which allows it to be a good conductor of electricity. ✓

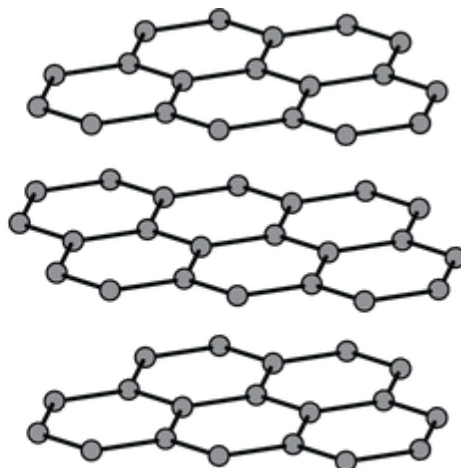


Unit 3 Progress Check: MCQ

6.



Diamond



Graphite

The structures of two allotropes of carbon are represented above. Which of the following statements best helps explain why diamond is much harder than graphite?

- (A) Diamond contains covalent bonds, whereas graphite contains ionic bonds.
- (B) Diamond contains ionic bonds, whereas graphite contains covalent bonds.
- (C) Carbon atoms in diamond have four covalent bonds, whereas graphite is made of layers that are held together by relatively weak dispersion forces. ✓
- (D) Carbon atoms in diamond have a sea of mobile electrons that make the structure strong, whereas graphite does not contain delocalized electrons.
7. A gaseous air-fuel mixture in a sealed car engine cylinder has an initial volume of **600. mL** at **1.0 atm**. To prepare for ignition of the fuel, a piston moves within the cylinder, reducing the volume of the air-fuel mixture to **50. mL** at constant temperature. Assuming ideal behavior, what is the new pressure of the air-fuel mixture?



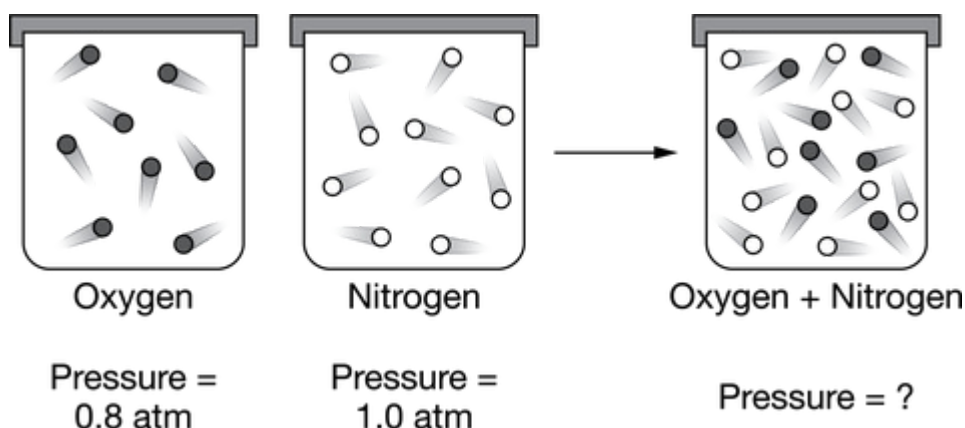
Unit 3 Progress Check: MCQ

- (A) About **0.083 atm**, because the initial volume of the gas mixture was 12 times larger than its final volume.
- (B) About **12 atm**, because the volume of the gas mixture decreased by a factor of 12. ✓
- (C) About **120. atm**, because the initial volume of the gas mixture was 120 times larger than its final volume.
- (D) About **550. atm**, because the change in the volume of the gas mixture was **550. mL**.
8. At **10. °C**, **20. g** of oxygen gas exerts a pressure of **2.1 atm** in a rigid, **7.0 L** cylinder. Assuming ideal behavior, if the temperature of the gas was raised to **40. °C**, which statement indicates the new pressure and explains why?
- (A) **1.9 atm**, because the pressure *P* decreases by the proportion $\frac{283}{313}$.
- (B) **2.3 atm**, because the pressure *P* increases by the proportion $\frac{313}{283}$. ✓
- (C) **0.52 atm**, because the pressure *P* decreases by the proportion $\frac{10.}{40.}$.
- (D) **8.4 atm**, because the pressure *P* increases by the proportion $\frac{40.}{10.}$.



Unit 3 Progress Check: MCQ

9.



Two sealed, rigid 5.0 L containers each contain a gas at the same temperature but at a different pressure, as shown above. Also shown are the results of transferring the entire contents of container 1 to container 2. No gases escape during the transfer. Assuming ideal behavior, which statement is correct regarding the total pressure of the gases after they are combined?

- (A) The total pressure of the gases in the mixture is the sum of the initial pressures of oxygen gas and nitrogen gas because pressure only depends on the total amount of gas when volume and temperature are held constant. ✓
- (B) The total pressure of the gases in the mixture is lower than the sum of the initial pressures of oxygen and nitrogen because some of the energy of the particles will be lost due to an increase in the number of collisions.
- (C) The total pressure of the gases in the mixture is higher than the sum of the initial pressures of oxygen and nitrogen because of the intermolecular forces that develop between oxygen and nitrogen molecules.
- (D) The total pressure of the gases in the mixture cannot be determined because the actual value of the temperature is not given.



Unit 3 Progress Check: MCQ

10.

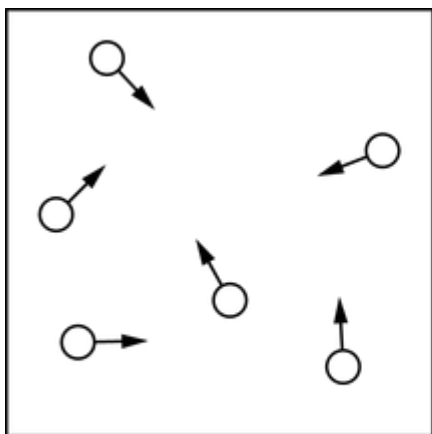


Diagram 1

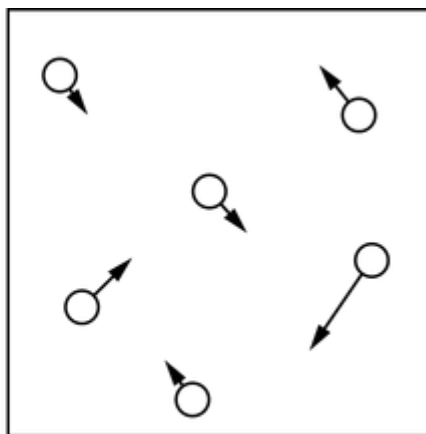


Diagram 2

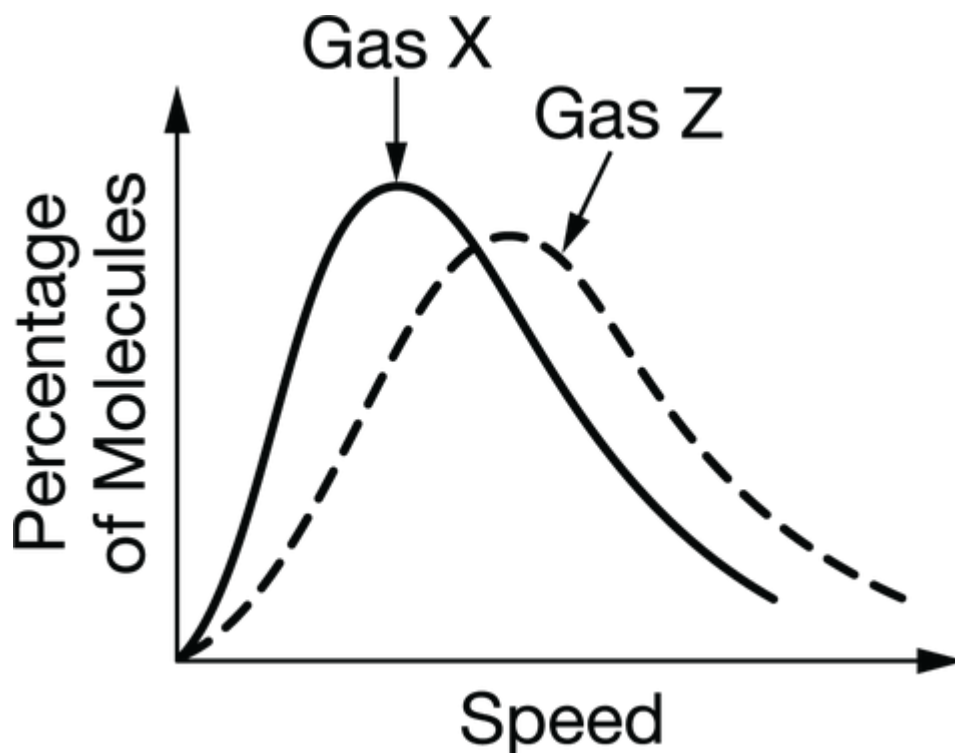
The diagrams above use arrows to represent the speed of a gas particle. Which of the diagrams best represents the speed of the particles of a gas at a fixed temperature, and why?

- (A) Diagram 1, because all the particles have the same speed.
- (B) Diagram 1, because the particles are moving in different directions.
- (C) Diagram 2, because the particles have a net kinetic energy of zero.
- (D) Diagram 2, because the particles have a variety of different speeds. ✓



Unit 3 Progress Check: MCQ

11.



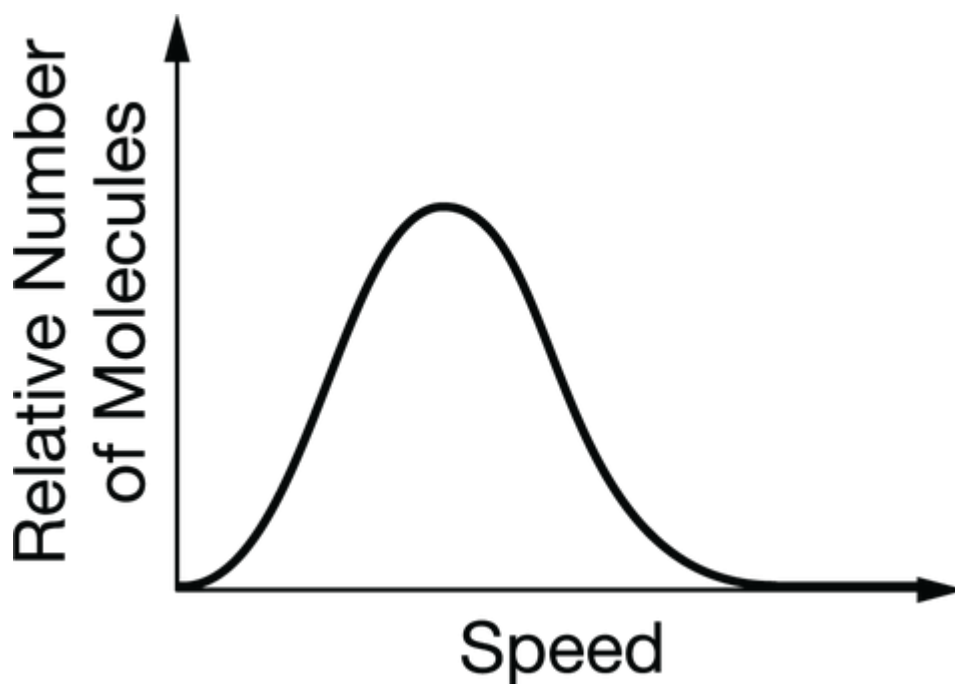
The two gas samples represented in the graph above are at the same temperature. Which of the following statements about the gases is correct?

- (A) The molecules of gas Z have a higher average kinetic energy than the molecules of gas X.
- (B) There are fewer molecules in the sample of gas Z than in the sample of gas X.
- (C) Gas Z has a smaller molar mass than gas X. ✓
- (D) Gas Z has a greater molar mass than gas X.

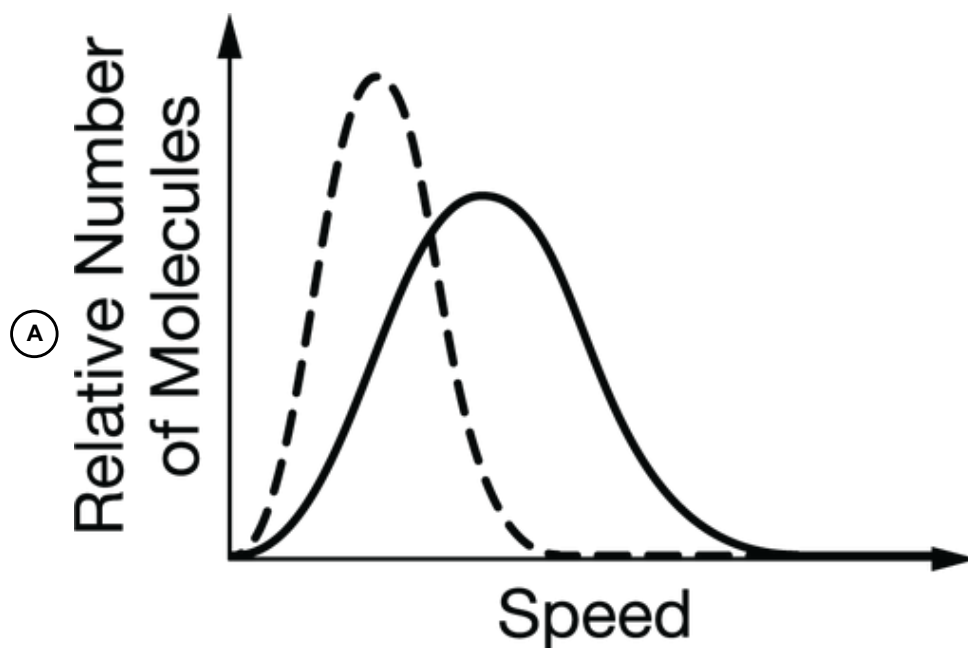


Unit 3 Progress Check: MCQ

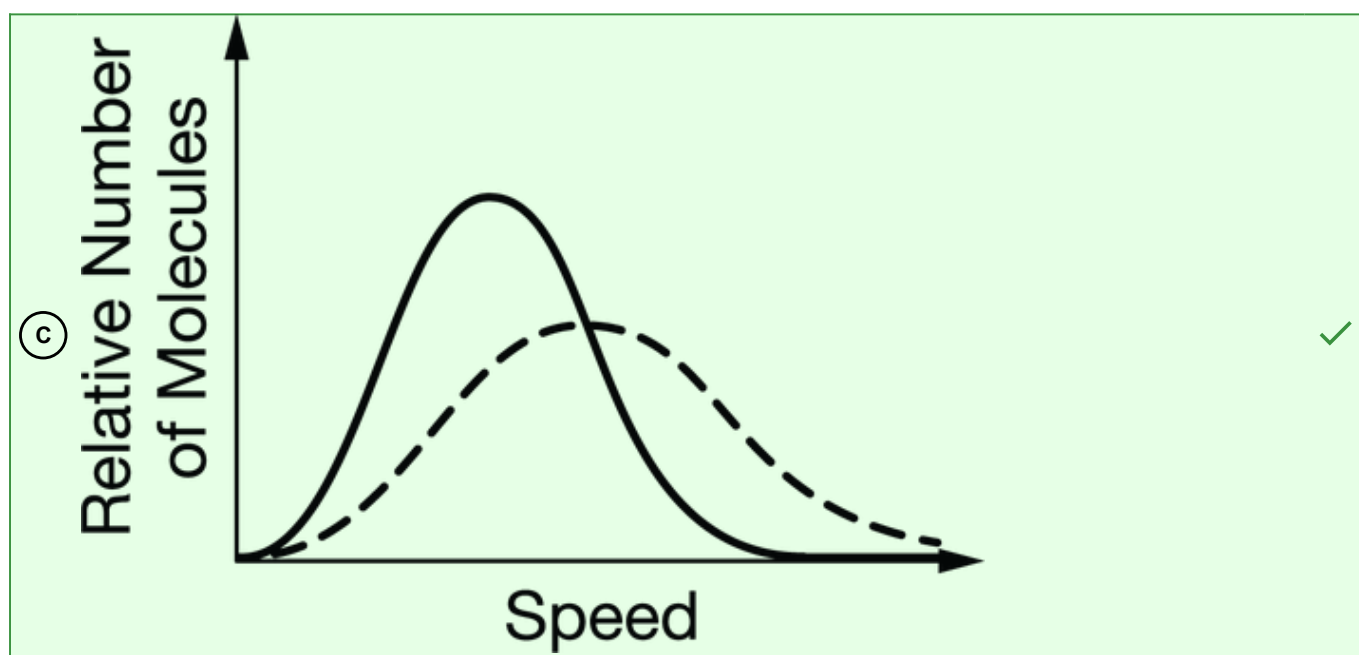
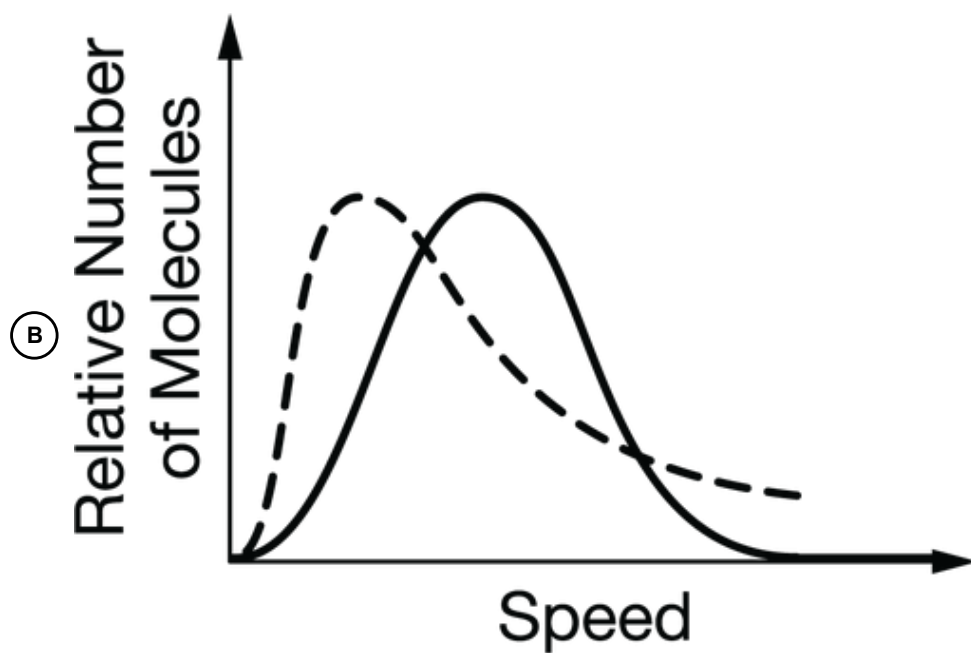
12.



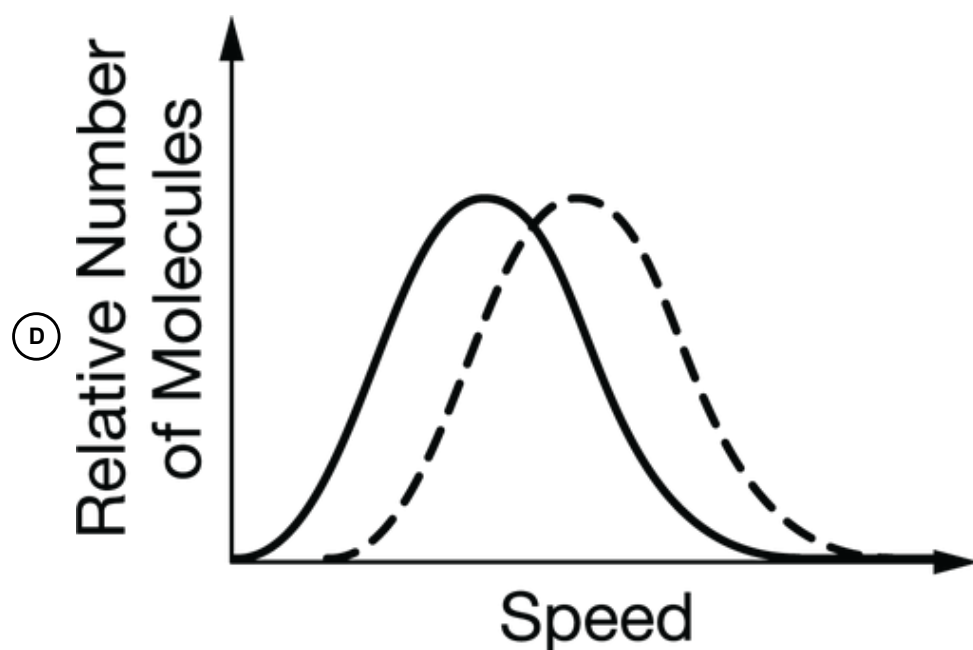
The diagram above shows the distribution of speeds for a sample of $\text{O}_2(g)$. Which of the following graphs shows the distribution of speeds for the same sample at a higher temperature (dashed line) ?



Unit 3 Progress Check: MCQ



Unit 3 Progress Check: MCQ



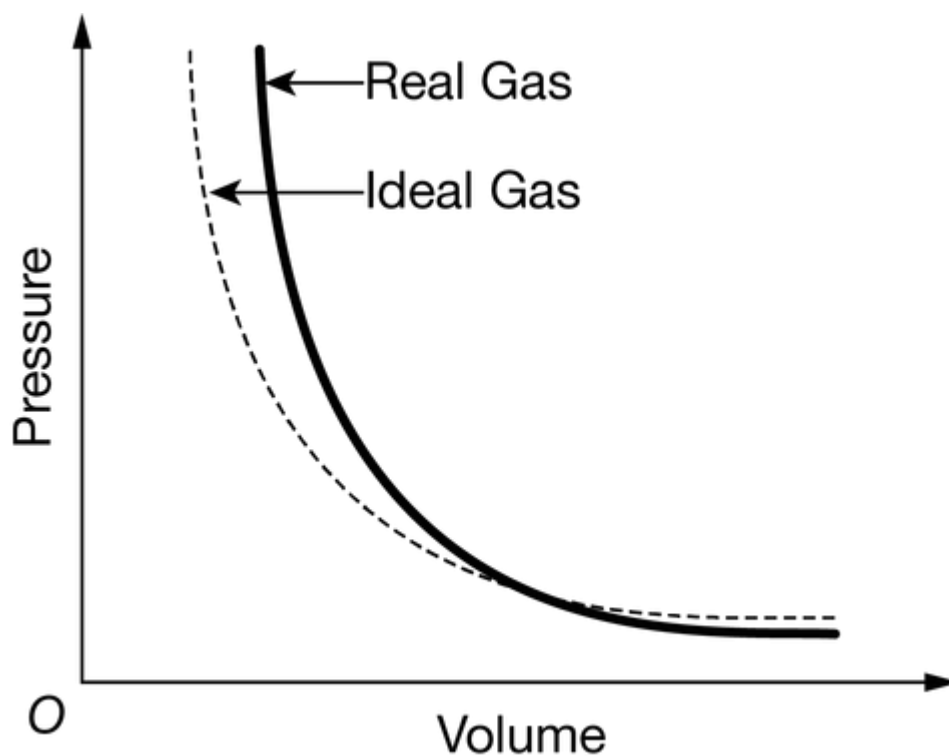
13. Equimolar samples of $\text{CH}_4(g)$ and $\text{C}_2\text{H}_6(g)$ are in identical containers at the same temperature. The $\text{C}_2\text{H}_6(g)$ deviates much more from ideal behavior than the $\text{CH}_4(g)$ does. Which of the following best helps explain this deviation?
- (A) C_2H_6 molecules have more hydrogen bonding than CH_4 molecules do.
- (B) C_2H_6 molecules have a larger, more polarizable electron cloud than CH_4 molecules do. ✓
- (C) The C_2H_6 molecules have a greater average kinetic energy than the CH_4 molecules have.
- (D) The C_2H_6 molecules have a greater average speed than the CH_4 molecules have.
14. The gases $\text{CO}_2(g)$ and $\text{NH}_3(g)$ can be liquefied at 20°C by compressing them to sufficiently high pressures. A student claims that $\text{NH}_3(g)$ can be liquefied at a lower pressure than $\text{CO}_2(g)$ can be liquefied. Which of the following is the best justification for this claim?



Unit 3 Progress Check: MCQ

- (A) At 20°C , the average speed of NH_3 molecules is greater than that of CO_2 molecules because NH_3 molecules have less mass than CO_2 molecules have.
- (B) CO_2 is a nonpolar molecule that has no significant intermolecular forces, whereas NH_3 has strong London dispersion intermolecular forces.
- (C) Both CO_2 and NH_3 are nonpolar molecules that have only London dispersion intermolecular forces, but the larger electron cloud of CO_2 molecules causes it to have stronger intermolecular forces.
- (D) CO_2 is a nonpolar molecule that has London dispersion intermolecular forces that are weaker than the dipole-dipole and London dispersion forces between the polar NH_3 molecules. ✓

15.



The graph above shows how a particular real gas deviates from ideal behavior at very high pressures. Based on this information, which of the following is most likely the gas and gives the reason based on kinetic molecular theory?



Unit 3 Progress Check: MCQ

- (A) H_2 , because it has the smallest mass.
- (B) N_2 , because its molecules have a triple bond.
- (C) Ne , because it has a completely filled valence shell.

(D) SO_2 , because it has the largest molecular volume. ✓

16. What volume of a 0.100 M HCl stock solution should be used to prepare 250.00 mL of 0.0250 M HCl ?

- (A) 1.00 mL
- (B) 16.0 mL

(C) 62.5 mL ✓

(D) $100.\text{ mL}$

17. A $500.\text{ mL}$ aqueous solution of Na_3PO_4 (molar mass = 164 g/mol) was prepared using 82 g of the solute. What is the molarity of Na_3PO_4 in the resulting solution?

- (A) 0.0010 M
- (B) 0.16 M
- (C) 0.25 M

(D) 1.0 M ✓



Unit 3 Progress Check: MCQ

18. How many grams of NaCl (molar mass = 58 g/mol) are needed to prepare 100. mL of a 0.25 M NaCl solution?

(A) 0.23 g

(B) 0.43 g

(C) 0.67 g

(D) 1.5 g



19.

Substance	Lewis Diagram	Molar Mass (g/mol)	Density (g/mL)	Boiling Point (K)
Methanol		32	0.79	338
Water		18	1.0	373

When methanol and water are mixed together, they form a homogeneous mixture. Based on the information in the table above, which of the following would be the best procedure for separating a mixture of methanol and water?



Unit 3 Progress Check: MCQ

(A) Filtration

(B) Distillation



(C) Paper chromatography

(D) Column chromatography

20. Which of the following methods is most appropriate to use to determine the number of different-colored components in a sample of black ink?

(A) Distillation at atmospheric pressure

(B) Elemental analysis to determine the mass ratio of C:H:N:O

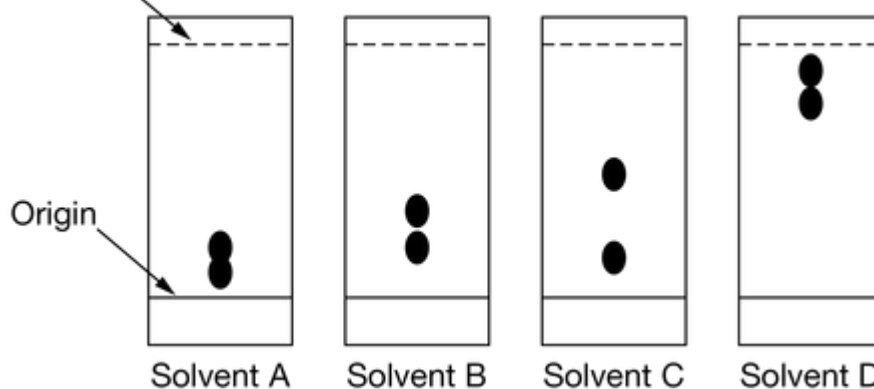
(C) Column chromatography using a nonpolar stationary phase and water as the mobile phase

(D) Paper chromatography using different solvents with a range of polarities as the mobile phase



Unit 3 Progress Check: MCQ

21. Solvent Front



The diagram above shows thin-layer chromatograms of the same mixture of two compounds. Based on the chromatograms, which solvent would be most effective at separating the two compounds if the same stationary phase is used for column chromatography?

(A) Solvent A

(B) Solvent B

(C) Solvent C

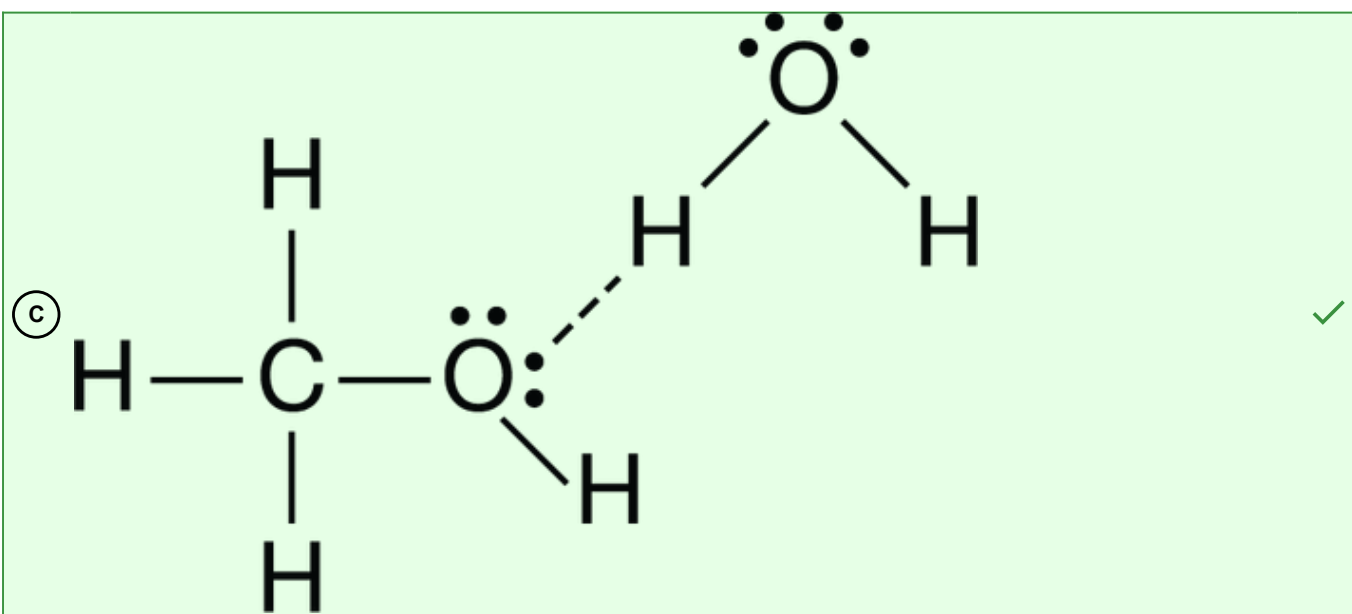
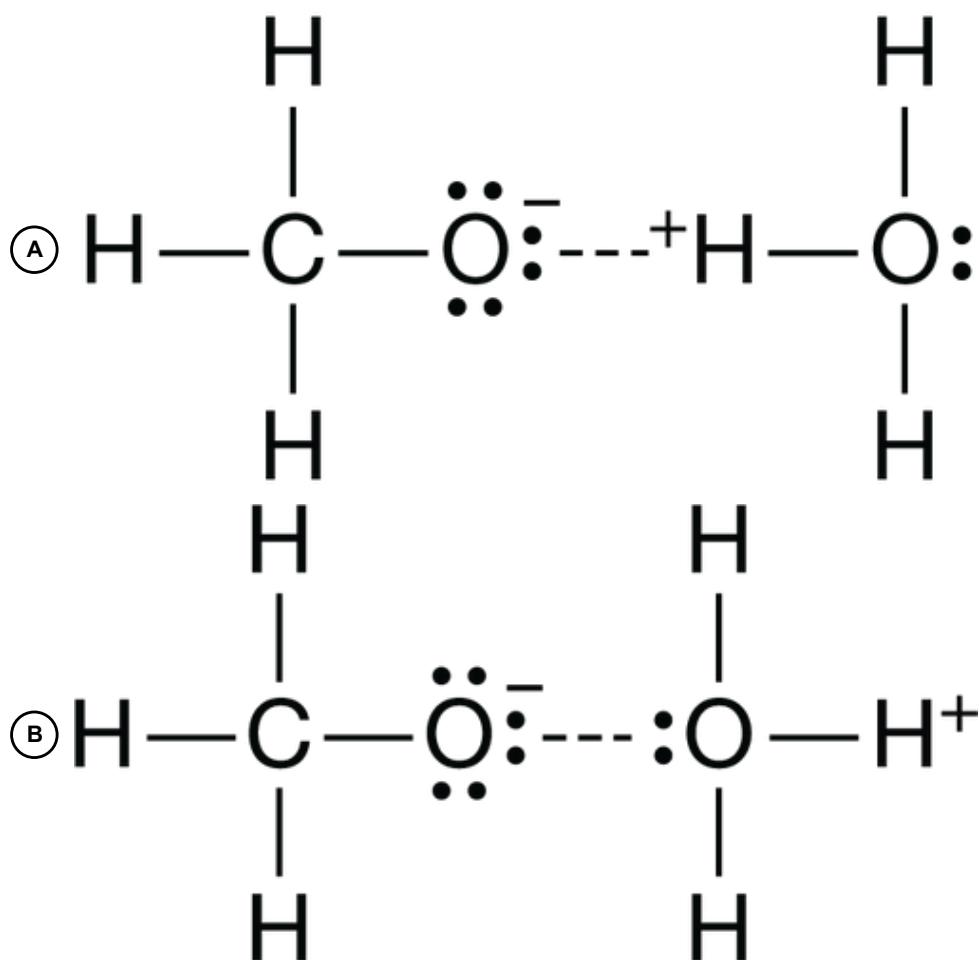


(D) Solvent D

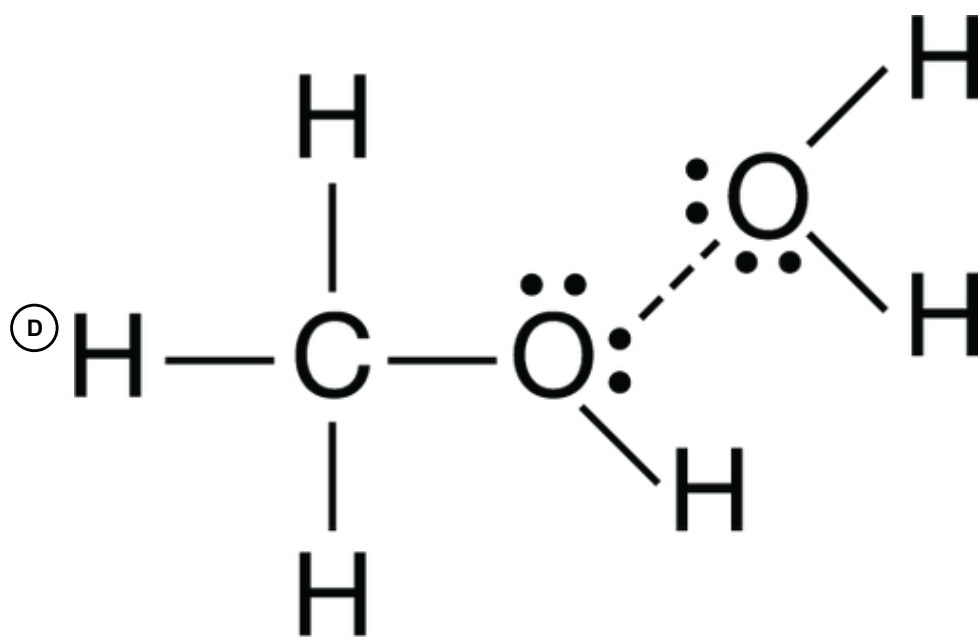
22. Methanol, CH_3OH , dissolves completely in water to form a solution that does not conduct electricity. Which of the following diagrams best shows the major type of attractive force that exists between the particles in the solution?



Unit 3 Progress Check: MCQ

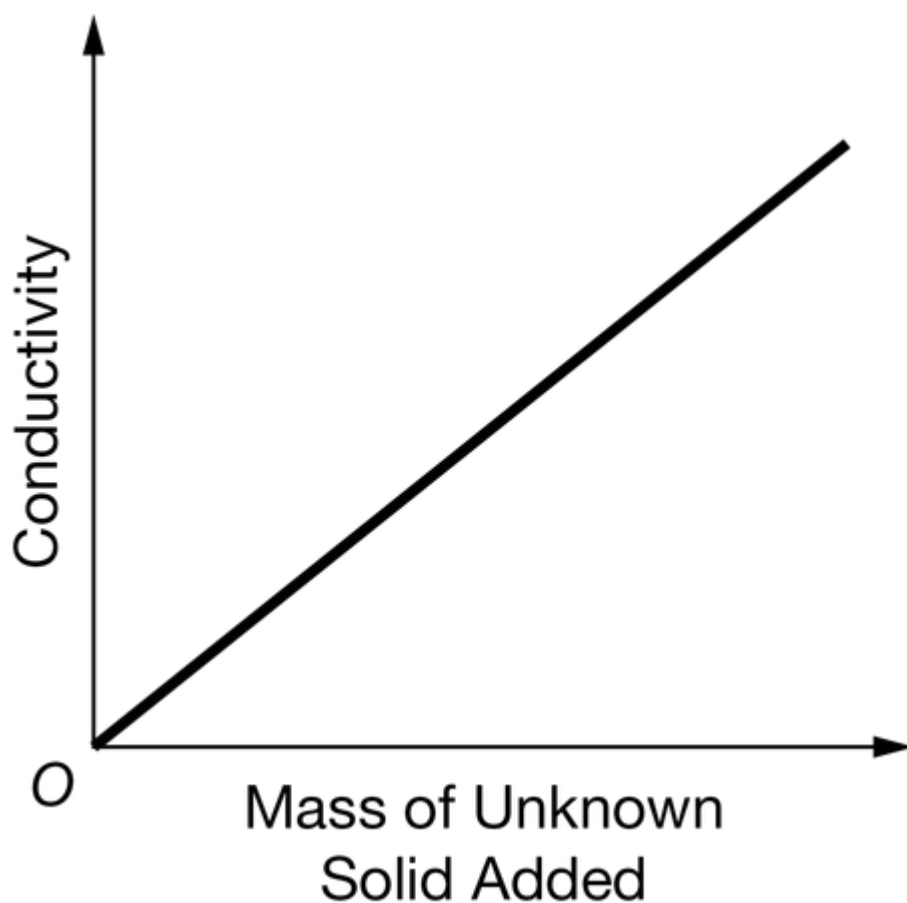


Unit 3 Progress Check: MCQ



Unit 3 Progress Check: MCQ

23.



A student adds a 1 g sample of an unknown, brittle solid to distilled water, stirs the mixture, and then measures its conductivity. The student repeats this procedure with more samples of the unknown solid and then produces the graph above. Which of the following statements about the graph and the properties of the solid is correct?

- (A) The increase in conductivity indicates that the unknown is an ionic solid that dissociates into ions when it dissolves in water. ✓
- (B) The increase in conductivity indicates that the unknown is a network covalent solid that ionizes in water.
- (C) The straight line plot indicates that the unknown solid is infinitely soluble in water.
- (D) The straight line plot indicates that the unknown solid has a very low solubility in water.



Unit 3 Progress Check: MCQ

24.

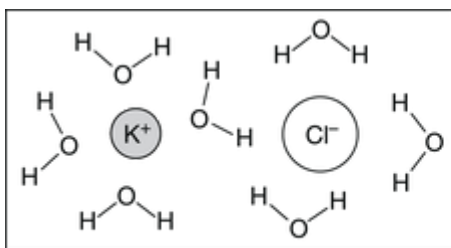


Diagram 1

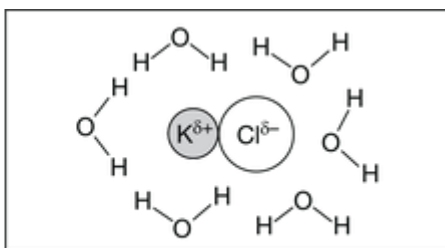


Diagram 2

Which of the diagrams above best represents the interactions that are responsible for the relatively large solubility of **KCl** crystals in water, and why?

- (A) Diagram 1, because strong ion-dipole interactions between **KCl** and water help to dissociate the solute. ✓
- (B) Diagram 1, because strong London dispersion forces between the **K⁺** and **Cl⁻** ions and water replace the weak London dispersion forces between two water molecules.
- (C) Diagram 2, because strong dipole-dipole forces between **KCl** and water help to separate the **KCl** units within the crystals.
- (D) Diagram 2, because the hydrogen bonds between water molecules expand to accommodate the **KCl** particles and pull them into solution.

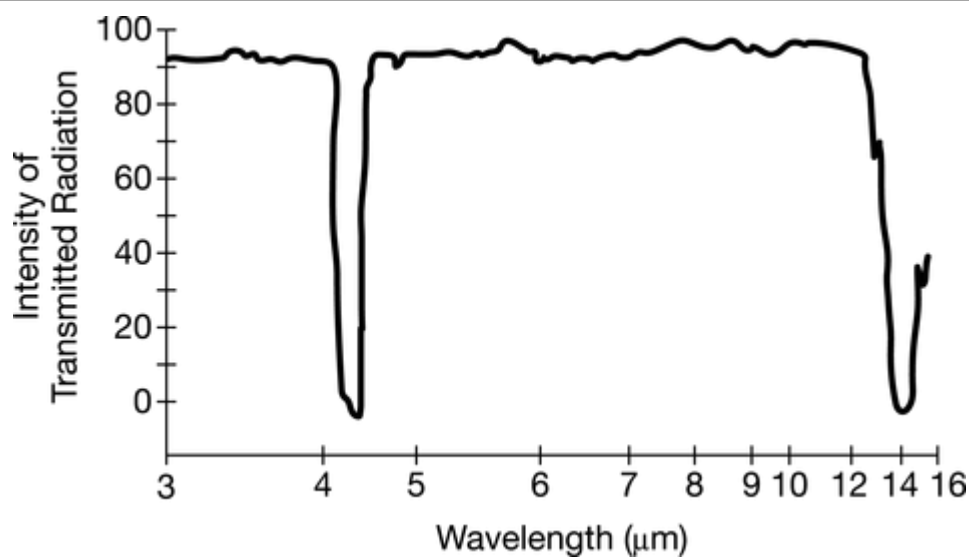
25. Which of the following best explains what happens as photons of visible light are absorbed by dye molecules?

- (A) Certain electrons in the dye molecule move to a higher energy level, with the difference in energy between the lower and higher energy levels being the same as the energy of the absorbed photons. ✓
- (B) Certain chemical bonds in the dye molecules begin to bend and stretch, with the difference in energy between the lower and higher vibrational states being the same as the energy of the absorbed photons.
- (C) The dye molecules begin to rotate faster in certain modes, with the difference in energy between the lower and higher rotational states being the same as the energy of the absorbed photons.
- (D) Certain covalent bonds in the dye molecules begin to break and re-form, with the bond energies of the bonds being the same as the energy of the absorbed photons.



Unit 3 Progress Check: MCQ

26.



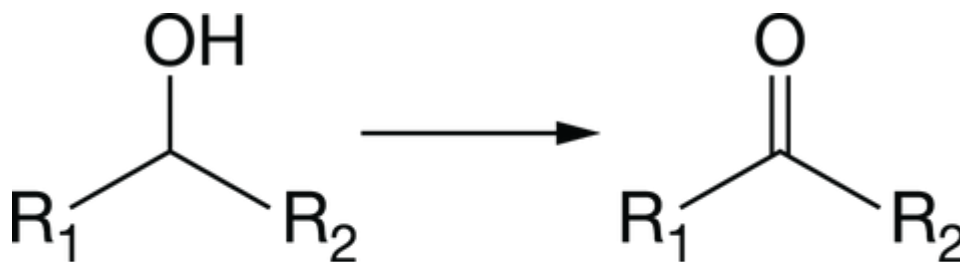
The infrared spectrum above represents the absorption of certain wavelengths of radiation by molecules of CO_2 . Which of the following best explains what occurs at the molecular level as the CO_2 molecules absorb photons of the infrared radiation?

- (A) The atoms in the CO_2 molecules increase their vibration as the bonds between the atoms bend and stretch. ✓
- (B) The molecules of CO_2 increase the energy of their rotational motions.
- (C) The electrons in the valence shells of the atoms in the CO_2 molecules are promoted to higher electronic energy levels.
- (D) The bonds between the atoms in the CO_2 molecules are continuously broken and then re-form.



Unit 3 Progress Check: MCQ

27.



Alcohol

Carbonyl

One type of organic molecule can be converted to another type of organic molecule through an oxidation-reduction process, as represented in the diagram above. Which of the following best explains why infrared spectroscopy is an appropriate method to confirm that the product contains a carbonyl?

- (A) The absorption of infrared radiation causes the nonbonding electrons on oxygen to transition into a higher electronic energy level.
- (B) The absorption of infrared radiation leads to an increase in molecular vibrational level associated with the stretching of the carbonyl. ✓
- (C) The absorption of infrared radiation causes the electrons in the pi bond to transition to a higher bonding energy level.
- (D) Infrared radiation leads to an increase in the molecular rotational level associated with the carbonyl as the axis of rotation.

28. Electromagnetic radiation with a maximum wavelength of **540 nm** ($5.4 \times 10^{-7} \text{ m}$) is needed for the study of the photoelectric effect in potassium atoms. What is the approximate frequency that corresponds to this wavelength?

- (A) $1.8 \times 10^{-15} \text{ s}^{-1}$
- (B) $1.6 \times 10^2 \text{ s}^{-1}$
- (C) $1.9 \times 10^6 \text{ s}^{-1}$
- (D) $5.6 \times 10^{14} \text{ s}^{-1}$ ✓



Unit 3 Progress Check: MCQ

29. In an experiment on the photoelectric effect, a photon with a frequency of $7 \times 10^{14} \text{ s}^{-1}$ was absorbed by a sample. What is the approximate energy of this photon?

(A) $5 \times 10^{-19} \text{ J}$



(B) $1 \times 10^{-15} \text{ J}$

(C) $2 \times 10^6 \text{ J}$

(D) $1 \times 10^{48} \text{ J}$

30.

Type of Electromagnetic Radiation	Frequency Range (Hz)	Energy Range (J)
Ultraviolet	$7.5 \times 10^{14} - 3 \times 10^{16}$	$5 \times 10^{-19} - 2 \times 10^{-17}$
Visible	?	?
Infrared	$3 \times 10^{11} - 4 \times 10^{14}$	$2 \times 10^{-22} - 3 \times 10^{-19}$

The frequency and energy ranges of photons in some parts of the electromagnetic spectrum are given in the table above. Which of the following could be the energy of a photon in the visible range?

(A) $9 \times 10^{-21} \text{ J}$

(B) $4 \times 10^{-20} \text{ J}$

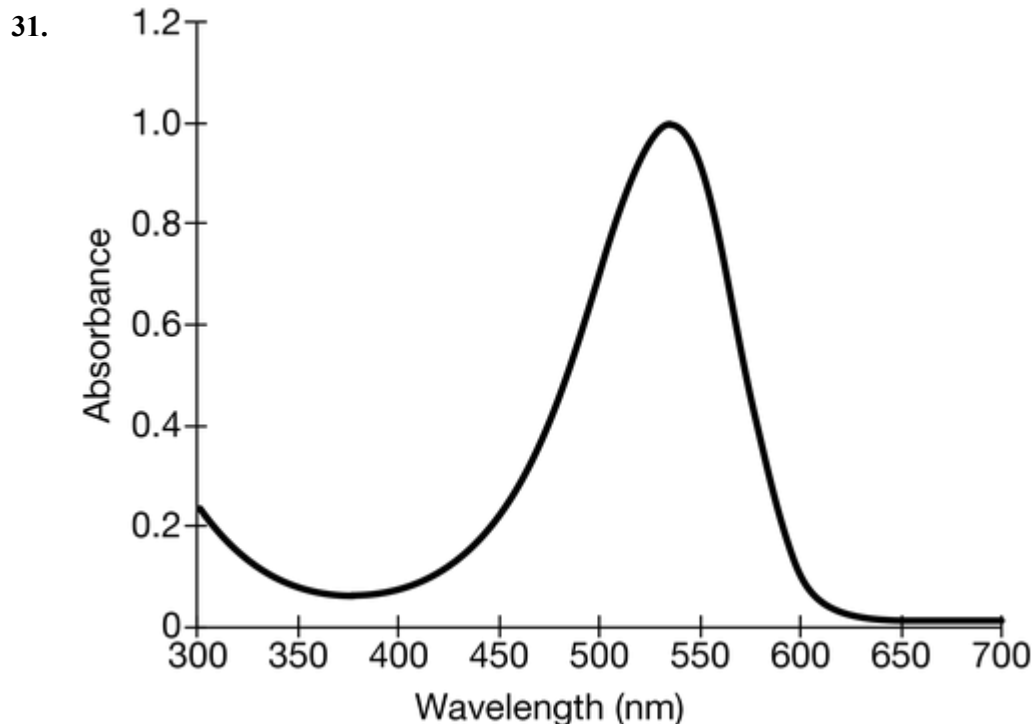
(C) $4 \times 10^{-19} \text{ J}$



(D) $9 \times 10^{-19} \text{ J}$



Unit 3 Progress Check: MCQ



The absorption spectrum of a certain red dye is shown above. If a student analyzing the same concentration of this dye neglected to wipe fingerprints off the cuvette before placing it in the spectrophotometer, how would the absorption curve be affected?

- (A) The peak of the curve would be higher because more light would be absorbed. ✓
- (B) The peak of curve would be lower because less light would be absorbed.
- (C) The peak of the curve would be shifted to the left because less light would be absorbed.
- (D) The peak of the curve would be shifted to the right because more light would be absorbed.

32. A student measures the absorbance of a solution containing FeSCN^{2+} ion using a spectrophotometer. The cuvette used by the student has two frosted walls and two transparent walls. The student properly orients the cuvette so that the path of the light goes through the transparent sides of the cuvette when calibrating the spectrophotometer. How will the measured absorbance of the FeSCN^{2+} be affected if the student incorrectly orients the cuvette so that the path of the light is through the frosted sides of the cuvette?



Unit 3 Progress Check: MCQ

- (A) The measured absorbance of the FeSCN^{2+} solution will not be affected.
- (B) The measured absorbance of the FeSCN^{2+} solution will be higher than the actual absorbance. ✓
- (C) The measured absorbance of the FeSCN^{2+} solution will be lower than the actual absorbance.
- (D) The effect on the measured absorbance of the FeSCN^{2+} solution depends on the concentration of the FeSCN^{2+} solution.
33. A student uses a spectrophotometer to analyze a solution of blue food dye. The student first rinses a cuvette with distilled water. Then the student adds the blue dye solution to the cuvette, forgetting to rinse the cuvette with the blue dye solution first. The student places the cuvette in the spectrophotometer and measures the absorbance of the solution. Assuming that some distilled water droplets were still in the cuvette when the blue dye solution was added, how would the measured absorbance be affected?
- (A) The measured absorbance would be too low, because the distilled water left in the cuvette would slightly dilute the solution. ✓
- (B) The measured absorbance would be too low, because the distilled water would change the optimum wavelength of absorbance.
- (C) The measured absorbance would be too high, because the water would allow more light to pass through the cuvette.
- (D) The measured absorbance would be too high, because the distilled water would also absorb some light.